JUAN GARCÍA NILA

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RESEARCH INTERESTS

Quantum error correction, non-Markovian open quantum systems, surface codes, bosonic: GKP/cat codes, continuous measurement and feedback, holonomic quantum computation and fault-tolerant architectures.

EDUCATION AND WORK EXPERIENCE

University of Southern California (USC) *May 2020 - August 2026*
Ph.D. in Electrical Engineering (Quantum Information Science) *GPA: 3.88, Los Angeles, CA, USA.*
Advisor: Todd A. Brun

University of Southern California (USC) *March 2020 - March 2024*
M.Sc. in Quantum Information Processing *GPA: 3.85, Los Angeles, CA, USA.*

Universidad Nacional Autónoma de México (UNAM) *February 2016 - January 2019*
M.Sc. in Physics *GPA: 9.83/10, Mexico City, Mexico*
Dissertation title: ”Bose–Einstein condensation in crystals with vacancies”
Advisor: M. A. Solís Atala

Universidad Nacional Autónoma de México (UNAM) *August 2011 - November 2015*
Bachelor of Science (Physics) *GPA: 9.97/10, Mexico City, Mexico.*
Top graduate of Class of 2012. (Gabino Barreda Medal)

RESEARCH EXPERIENCE

Research in Quantum Information at LANL *June 2026- August 2026*

- An immersive 10-week curriculum designed around quantum computing. Projects on D-Wave Systems, Rigetti, Quantinuum and IBM.
- Research about continuous quantum error correction. Under the supervision of Dr Lucas Jhons and Luis Pedro Garcia Pintos.

Research in Quantum Information Science *Los Angeles, CA, USA. March 2020- August 2026*

- Developed continuous-measurement–based holonomic quantum gates for fault-tolerant quantum computation.
- Derived and analyzed the performance of different Markovian and non-Markovian master equations (Nakajima–Zwanzig, Time Convolutionless Lindblad TCL on a spin in a leaky cavity model which can be solved analytically.
- Studied quantum error correction under realistic noise models including correlated and non-Markovian environments.

Theoretical research in Many-Body Quantum Theory *Mexico City, Mexico. March 2015- Nov 2019*

- Studied analytically the effects on the thermodynamic behavior of an ideal boson gas immersed in an imperfect crystal modeled by a Dirac Kronig-Penney potential.

Experimental research in the Applied Optics Lab *Mexico City, Mexico. August 2016 - March 2017*

- Studied optical traps, manipulating microscopical objects using structured light in Pedro Quinto’s lab.

Mitacs Globalink Research Intern (Funded Program) *Burnaby, BC, Canada. May 2016 - June 2016*

- Implemented an electronic feedforward system on a vortex tube to prevent the overheating of the equipment.
- Studied the spin transport in a trapped ultra-cold ^{87}Rb gas at temperatures above quantum degeneracy.

PUBLICATIONS

Published

1. **J. García-Nila** and T. Brun. Continuous quantum correction on Markovian and non-Markovian models" (Feb 2026). Phys. Rev. A 113, 022442.

2. Z. Xia, **J. Garcia-Nila** and D. Lidar. "Markovian and non-Markovian master equations versus an exactly solvable model of a qubit in a cavity" (2024). Phys. Rev. Applied 22, 014028

3. J. G. Martínez-Herrera, **J. García-Nila** and M. A. Solís. (2019) "Bose Gas with generalized dispersion relation plus an energy gap", Physica Scrypta, Vol 94, Num 7.

Preprints

4. **J. Garcia-Nila**, A. Lanka, T A Brun. *Holonomic quantum gates via continuous measurement in bosonic codes: GKP and cat states.* arXiv2608.11369

5. A. Lanka, **J. Garcia-Nila**, T A Brun. *Continuous measurement-based holonomic quantum computation.* arXiv2510.06725

6. A. Lanka, **J. Garcia-Nila**, T A Brun. *Steering paths mid-flight for fault-tolerance in measurement-based holonomic gates* March 2026 arXiv2603.02552

TEACHING EXPERIENCE

Teaching Assistant *Jan 2023- Present*
University of Southern California
-Courses in the Master of Quantum Information Science:

Introduction to Quantum Information Science (EE520) and Quantum Information Theory (EE599). Dr Todd A Brun

Open Quantum Systems (PHYS 550) Dr Daniel A Lidar

Linear Algebra for Engineering (EE510). Dr. Olaoluwa Adigun and Dr. Ketan Savla.

Associate Professor *May 2019- August 2021*
Facultad de Ciencias UNAM, Mexico City, Mexico
Contemporary Physics (Fall 2019), Basic Science (Fall 2020),Electromagnetism(Spring 2021) and Introduction to Differential and Integral on several variables (Spring 2019).

PRESENTATIONS, SCHOOLS AND INVITED TALKS

Invited Talks

Seminario de Investigacion Semanal *Instituto Mexicano del Petroleo, Mexico City. May 28 2025*
Seminario de investigacion Sandoval Vallarta *Instituto de Fisica UNAM, Mexico City. May 30 2025*
-Invited Talk. "Introduction to Quantum Advantage and Error Correction".

APS March Meeting
2025 *March 18, Anaheim CA.*
2024 *March 3, Minneapolis MN, USA.*
2023 *March 9, Las Vegas NV, USA.*

Poster Presentations

QuAMP 2025 *Newcastle University, UK. Sep 2 2025*
SQUINT 2025 *University of New Mexico, Albuquerque NM. Oct 9 2025*
Markovian and non-Markovian master equations versus an exactly solvable model of a qubit in a cavity

Schools

The Schrödinger Sessions *University of Maryland. July 31 2025*
Quantum Ideas Summer School *Duke Quantum Center, NC. June 9-13 2025*
US Quantum Information Science Summer School *Oak Ridge TN, USA. July 18 2024*

LANGUAGES

Spanish: Native Speaker
English: TOEFL iBT 105. Reading: 29, Listening: 25, Speaking: 25 and Writing: 26.

TECHNICAL SKILLS

Programming: Python, Julia, Mathematica, Octave, MATLAB & LaTeX
GRE: Quantitative: 161, Verbal: 152